

Question			Marking details	Marks available					Prac
				AO1	AO2	AO3	Total	Maths	
3	(a)	(i)	Sirius A: $\lambda_{\text{max}} = 290 \times 10^{-9} \text{ [m]}$ and Canopus: $\lambda_{\text{max}} = 400 \times 10^{-9} \text{ [m]}$ [$\pm 10 \text{ nm}$] for both (1) Attempt at applying $\lambda_{\text{max}} T = 0.0029$ to both stars, even if powers of 10 incorrect (1) No need to change unit of λ to m Correct application, either by confirming Wien constant or star temperatures or λ_{max} (1) Accept correct calculations of λ_{max} even if no reference is made to the graph Application of Wien's law to one star only award 1 mark only			3	3	2	
		(ii)	Sirius A (1) Greater spectral intensity (at 'blue' end or at shorter wavelengths or towards 400 nm) (1) Don't accept peak wavelength of Sirius A is closest to the blue end of the spectrum than Canopus. Don't accept reference to temperature by itself e.g. Sirius has a higher temperature so therefore must be bluer. Accept ratio of $\frac{B}{R}$ for both stars, e.g. $\frac{6.1}{1.7}$ against $\frac{1.6}{0.8}$		1	1	2		

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	(b)	(i)	Correct use of $P = A\sigma T^4$ for either Sirius A or Canopus or both i.e. Canopus: $P = 4 \times \pi \times (4.97 \times 10^{10})^2 \times 5.67 \times 10^{-8} \times (7\,250)^4$ Sirius A: $P = 4 \times \pi \times (1.19 \times 10^9)^2 \times 5.67 \times 10^{-8} \times (10\,000)^4$ (1) Canopus: $P = 4.9 \times 10^{30}$ [W] (1) Sirius A: $P = 1.0 \times 10^{28}$ [W] (1) Accept powers of 10 error in answers omission of 4 deduct 1 mark $\frac{4.9 \times 10^{30}}{1.0 \times 10^{28}}$ (≈ 500) shown (1) Alternative: Attempt at $P = A\sigma T^4$ used in ratio (1) e.g. $\frac{L_C}{L_S} = \frac{49.7 \times 7\,250^4}{1.192 \times 10\,000^4} \approx 500$ (1)	1	1 1		4	
		(ii)	$I = \frac{1.0 \times 10^{28}}{4\pi (8.15 \times 10^{16})^2}$ substitution (1) ecf $I = 1.19 \times 10^{-7}$ [W m ⁻²] (1) Accept 1.21×10^{-7} [W m ⁻²]	1	1		2	2
		(iii)	Canopus is further away from earth or Sirius A is closer to earth (1) Intensity reaching earth $\propto \frac{1}{R^2}$ or P from star spread out over greater surface area (1) Accept intensity equation Don't accept intensity $\propto \frac{1}{\text{distance}}$ or because of the inverse square law		2		2	
			Question 3 total	2	7	4	13	8 0